

| Questions |     |  | Answers                                                                                                                                                       | Marks |
|-----------|-----|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 2         | (a) |  | Using $v^2 = u^2 + 2as$ , $u = 0$ or $\frac{1}{2}mv^2 = mg\Delta h$ to arrive at<br>$v = \sqrt{2 \times 9.81 \times 1.96}$<br>$v = 6.20(12) \text{ m s}^{-1}$ | M1    |
|           |     |  |                                                                                                                                                               | A1    |
|           | (b) |  | unchanged                                                                                                                                                     | B1    |
|           |     |  | Normal contact force on ball is vertical                                                                                                                      | B1    |
|           | (c) |  | $v^2 = u^2 + 2as$ , $v = 0$ or $\frac{1}{2}mv^2 = mg\Delta h$ to obtain rebound speed<br>$v = \sqrt{2 \times 9.81 \times 0.98} = 4.38 \text{ (49) m s}^{-1}$  | B1    |
|           |     |  | impulse = change in momentum $m(v_f - v_1)$<br>impulse = $0.034(4.3849 - (-6.2012)) = 0.36 \text{ N s}$ or $\text{kg m s}^{-1}$                               | A1    |

